Simple Shopping Bill with Discount

1. Problem Statement

Calculate the total cost of a product using its price and quantity. Give a 10% discount when the total amount is ₹1,000 or more.

2. Objectives

1. Read product name, price, and quantity.
2. Calculate the total amount.
3. Apply discount using an if-else condition.
4. Print the final bill.

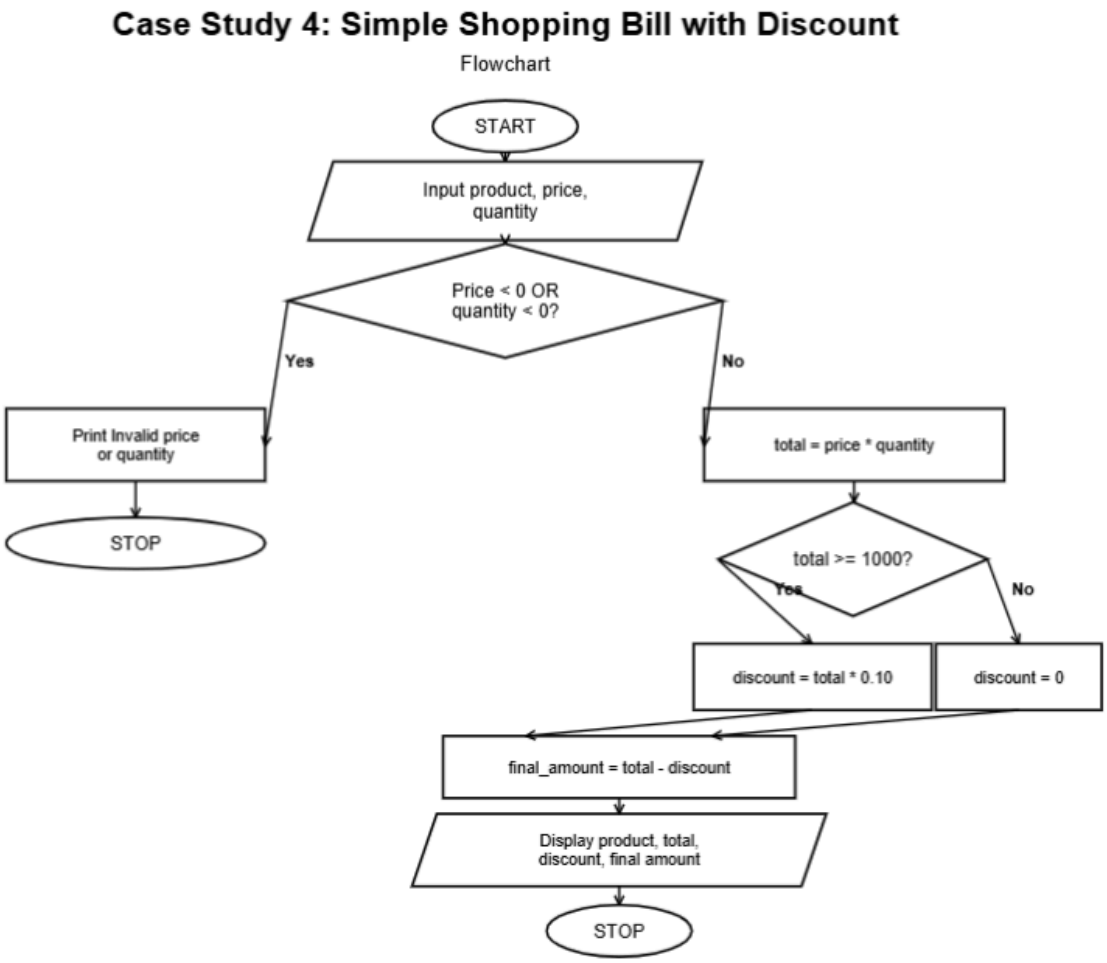
3. Assumptions

1. Price and quantity cannot be negative.
2. Discount is 10% when the total is at least ₹1,000.
3. No tax or delivery charge is included.

4. Algorithm

1. Start.
2. Read product name, price, and quantity.
3. Calculate `total = price × quantity`.
4. If total is at least 1,000, calculate a 10% discount; otherwise, discount is zero.
5. Calculate `final_amount = total - discount`.
6. Display the bill.
7. Stop.

5. Flowchart



None

```
    |
```

6. Pseudocode

None

```
INPUT product, price, quantity

IF price < 0 OR quantity < 0 THEN
    DISPLAY "Invalid price or quantity"
```

```
ELSE
    total = price * quantity

    IF total >= 1000 THEN
        discount = total * 0.10
    ELSE
        discount = 0
    ENDIF

    final_amount = total - discount

    DISPLAY total, discount, final_amount
ENDIF
```

8. Test Case and Expected Output

Input:

```
None
Product = Bag
Price = Rs. 600
Quantity = 2
```

Calculation:

- Total Amount = Price × Quantity
- Total Amount = 600 × 2 = ₹1,200
- Discount = 10% of ₹1,200 = ₹120
- Final Amount = ₹1,200 − ₹120 = ₹1,080

Expected Output:

```
None
Enter product name: Bag
Enter price: Rs. 600
Enter quantity: 2

----- SHOPPING BILL -----
```

```
Product      : Bag
Total amount : Rs. 1200.0
Discount     : Rs. 120.00
Final amount : Rs. 1080.00
-------------------------
```

9. Conclusion

The program calculates a shopping bill and applies a discount when the purchase reaches the specified amount. It uses conditional statements to check whether the total amount is ₹1,000 or more and calculates the final amount after discount.